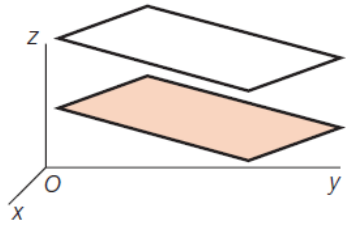
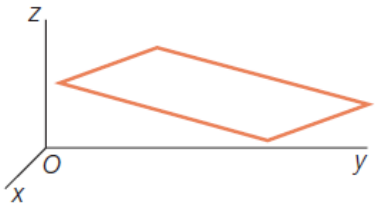
Intersection of Two Planes

Two planes can intersect in the following ways:



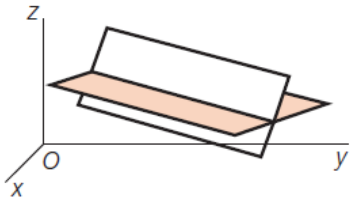
Planes are _____

_____ Solutions



Planes are _____

_____ Solution



Planes _____

_____ Solution

Example 1: Determine the intersection of the planes

$$x + 4y - 3z = 6$$
$$2x + 8y - 6z = -11$$

Challenge: What's the **ONE** change we can make to the system above so that there will be infinite solutions?

a) Coincident Planes?

b) Intersecting Planes?

Example 2: Determine the intersection of the planes

$$2x + 3y + 4z = 10$$
$$3x + y - z = 1$$

Example 3: Determine the angle of intersection between the two planes.

$$2x + 3y + 4z = 10$$
$$3x + y - z = 1$$

Recall:
1. Planes are parallel if:

2. Planes are perpendicular if:

3. Angle between intersecting planes can be calculated using: